PHYSICS

Paper 1 (Theory)

Maximum Marks: 70 **Time Allowed: Three Hours**

Reading Time: Additional Fifteen Minutes

Instructions to Candidates

1. You are allowed an additional fifteen minutes for only reading the question paper.
2. You must NOT start writing during reading time.
3. This question paper has 9 printed pages.
4. It has twenty questions in this paper. Answer all questions.
5. There are four sections in the paper: A, B, C and D. Internal choices have been provided in two questions each in Sections B, C and D.
6. Section A consists of one question having fourteen subparts of one mark each.
7. While attempting Multiple Choice Questions in Section A, you are required to write only ONE option as the answer.
8. Section B consists of seven questions of two marks each.
9. Section C consists of nine questions of three marks each.
10. Section D consists of three questions of five marks each.
11. The intended marks for questions are given in brackets [ ].
12. A list of useful constants and relations is given below these instructions.
13. A simple scientific calculator without a programmable memory may be used for calculations.

USEFUL CONSTANTS AND RELATIONS

1.	Planck's constant	h	=	$6{\cdot}6 \times 10^{-34}$ J s
2.	Speed of light in vacuum	c	=	3×10^{8} m s^{-1}
3.	Charge of a proton	e	=	$1{\cdot}6 \times 10^{-19}$ C
4.	Mass of an electron	m_e	=	$9{\cdot}1 \times 10^{-31}$ kg
5.	Permittivity of vacuum	ε_0	=	$8{\cdot}85 \times 10^{-12}$ F m^{-1}
6.	Permeability of vacuum	μ_0	=	$4\pi \times 10^{-7}$ H m^{-1}
7.	One atomic mass unit	1 u	=	$1{\cdot}66 \times 10^{-27}$ kg = 931·5 MeV/c^2
8.	One electron volt	1 eV	=	$1{\cdot}6 \times 10^{-19}$ J
9.		1 MeV	=	$1{\cdot}6 \times 10^{-13}$ J
10.	Rydberg constant	R	=	$1{\cdot}097 \times 10^{7}$ m^{-1}

Note: The questions in this sample paper follow the format of the ISC Specimen Question Paper 2027. The weightage allocated to each unit in the syllabus has been strictly adhered to.

SECTION A – 14 MARKS

Question 1

(A) In questions (i) to (vii), choose the correct alternative (a), (b), (c) or (d) for each of the questions given below.

(i) Point charges of +30 nC, −10 nC and +5 nC lie inside a large closed spherical surface, and a further charge of +40 nC lies outside it. The net electric flux through the surface is: *(Apply)* [1]

(a) $5{\cdot}1 \times 10^3$ N m^2 C^{-1}

(b) $2{\cdot}8 \times 10^3$ N m^2 C^{-1}

(c) $7{\cdot}3 \times 10^3$ N m^2 C^{-1}

(d) zero

(ii) When the current in a coil changes from 9 A to 3 A in 4×10^{-2} s, the emf induced in the coil is 3 V. The self-inductance of the coil, in millihenry, is: *(Apply)* [1]

(a) 5

(b) 10

(c) 20

(d) 40

(iii) Select the odd one out from the following options. *(Analyse)* [1]

(a) Aluminium

(b) Sodium

(c) Calcium

(d) Bismuth

(iv) Match the entries in Column I with those in Column II. *(Recall)* [1]

Column I	Column II
1. Slope of the V_s–ν graph	A. $-\varphi/e$
2. Intercept of the V_s–ν graph on the ν-axis	B. Threshold frequency ν_0
3. Intercept of the V_s–ν graph on the V_s-axis	C. h/e
4. de Broglie wavelength of an electron accelerated through V volt	D. $1{\cdot}227/\sqrt{V}$ nm

(a) 1-C, 2-B, 3-A, 4-D

(b) 1-B, 2-C, 3-A, 4-D

(c) 1-C, 2-A, 3-B, 4-D

(d) 1-A, 2-B, 3-C, 4-D

(v) Identify the correct statements about an n-type semiconductor. *(Understand)* [1]

(P) Electrons are the majority charge carriers.

(Q) It is obtained by doping with a pentavalent impurity.

(R) The crystal carries a net negative charge.

(S) The concentration of holes is greater than that of electrons.

(a) Both (P) and (S)

(b) Both (P) and (Q)

(c) Both (Q) and (R)

(d) Both (R) and (S)

(vi) Given below are two statements marked, Assertion and Reason. Read the two statements and choose the correct option. [1]

Assertion: The graph of v against u for the real images formed by a concave mirror lies in the third quadrant.

Reason: For a real image formed by a concave mirror, both u and v are negative in the Cartesian sign convention. *(Analyse)*

(a) Both Assertion and Reason are true, and Reason is the correct explanation for Assertion.

(b) Both Assertion and Reason are true, but Reason is not the correct explanation for Assertion.

(c) Assertion is true and Reason is false.

(d) Assertion is false and Reason is true.

(vii) A biconvex lens of glass of refractive index 1·5 is dipped in a transparent liquid and is found to behave like a diverging lens. The refractive index of the liquid could be: *(Apply)* [1]

(a) 1·00

(b) 1·33

(c) 1·65

(d) 1·50

(B) Answer the following questions briefly:

(i) In a potentiometer experiment, the current in the potentiometer wire is doubled by adjusting the rheostat. How does the balance length for a given cell change? *(Analyse)* [1]

(ii) By what factor does the magnetic field at the centre of a circular coil change if the current is tripled, the number of turns is doubled and the radius is doubled? *(Apply)* [1]

(iii) State one advantage of a reflecting telescope over a refracting telescope. *(Recall)* [1]

(iv) Fill in the blank with an appropriate term from the bracket. [1]

If the separation of the two slits in Young's double slit experiment is doubled, the fringe width _____ (is halved, is doubled, remains the same). *(Understand)*

(v) In an experiment on the photoelectric effect, the slope of the graph of cut-off voltage against frequency of the incident light is found to be $4{\cdot}1 \times 10^{-15}$ V s. Calculate the value of Planck's constant. *(Apply)* [1]

(vi) A gamma-ray photon of energy 2·0 MeV produces an electron–positron pair. Calculate the total kinetic energy of the pair. (Take the rest energy of an electron as $m_e c^2$.) *(Apply)* [1]

(vii) Explain why the resistance of a semiconductor decreases when its temperature is raised. *(Understand)* [1]

SECTION B – 14 MARKS

Question 2

(i) Figure 1 shows a closed surface S and four point charges. Calculate the electric flux emanating from the closed surface S. *(Apply)* [2]

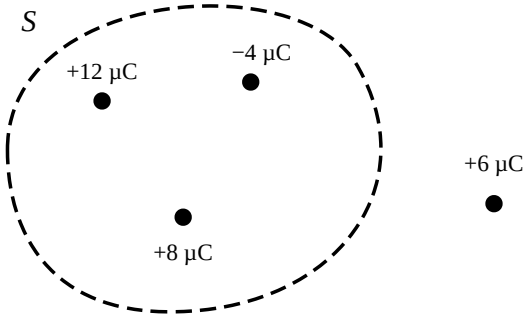


Figure 1

OR

(ii) The electric potential V varies with distance x as shown in Figure 2. Construct a graph of the electric field strength E as a function of x. *(Analyse)* [2]

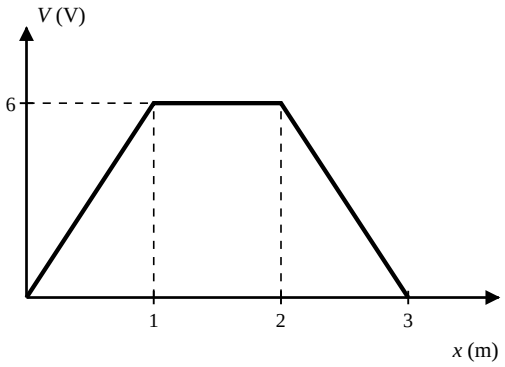


Figure 2

Question 3

Three identical capacitors, each of capacitance 6 µF, are connected in parallel across a 12 V battery. (a) Calculate the energy stored in the system. (b) Find the ratio C_s/C_p of the equivalent capacitance when the three are connected in series to that when they are connected in parallel. *(Apply)* [2]

Question 4

(i) The terminal potential difference *V* of a cell is measured for different currents *I* drawn from it, and plotted in Figure 3. Use the graph to find (a) the emf and (b) the internal resistance of the cell. *(Apply)* [2]

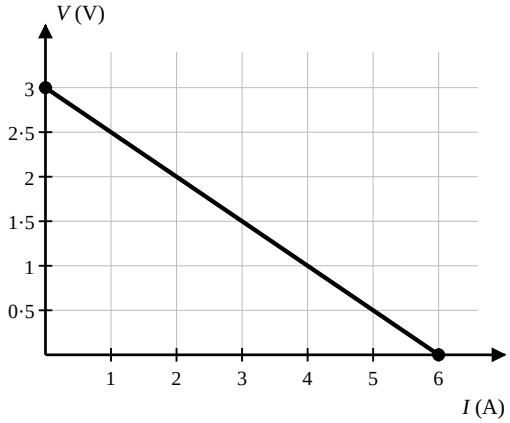


Figure 3

OR

(ii) A heater takes 20 minutes to boil a given amount of water. Its coil is cut and one-fourth of its length is used in the heater. How much time will it now take to boil the same amount of water using the same source, starting from the same initial temperature? *(Apply)* [2]

Question 5

(a) State the SI unit of the magnetic dipole moment of a current-carrying loop and the direction of the moment. (b) Calculate the magnetic moment of a circular coil of 100 turns and radius 5 cm carrying a current of 2 A. *(Apply)* [2]

Question 6

(i) A convex lens of focal length 25 cm is in contact with a concave lens of focal length 40 cm. Calculate the power and the focal length of the combination. *(Apply)* [1]

(ii) The angle of a glass prism is 60° and the angle of minimum deviation is 38°. Without using the refractive index of the glass, find the angle of incidence of the ray suffering minimum deviation. *(Apply)* [1]

Question 7

(i) Arrange the following electromagnetic waves in the increasing order of their frequencies: X-rays, microwaves, ultraviolet rays, infrared rays. *(Recall)* [1]

(ii) Calculate the frequency of an electromagnetic wave of wavelength 15 nm. *(Apply)* [1]

Question 8

The work functions of two metals M_1 and M_2 are 2·2 eV and 4·5 eV respectively. Monochromatic light of wavelength 500 nm falls on each. Perform the necessary calculations to find which metal emits photoelectrons, and the maximum kinetic energy of the ejected electrons. *(Apply)* [2]

SECTION C – 27 MARKS

Question 9

(i) Show that the electric field at a point on the axial line of a short electric dipole, at a distance r from its centre, is given by $E = \frac{1}{4\pi\varepsilon_0}\frac{2pr}{(r^2 - l^2)^2}$, where the symbols have their usual meaning. *(Understand)* [3]

OR

(ii) An infinite plane sheet carries a uniform surface charge density +σ. Using Gauss's law, find the electric field at a point at a distance r from it. Hence write the resultant field at a point **between** two such parallel sheets carrying +σ and −σ, and give its direction. *(Understand)* [3]

Question 10

(i) In the network of Figure 4 the cells have negligible internal resistance and the current through the 4 Ω resistor is 1·5 A. Using Kirchhoff's laws, find (a) the current I_1, (b) the current I_2 and the emf E of the second cell. *(Apply)* [3]

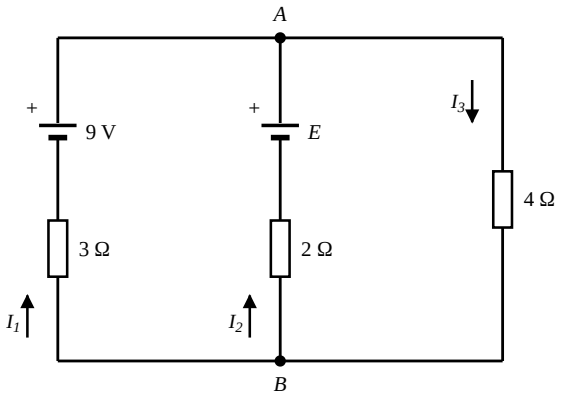


Figure 4

OR

(ii) In the metre bridge of Figure 5, the resistance in the left gap is 2 Ω and the null point is found at 40 cm from the left end. What resistance should be connected to the unknown resistance X, and how, so that the null point shifts to 50 cm from the left end? *(Analyse)* [3]

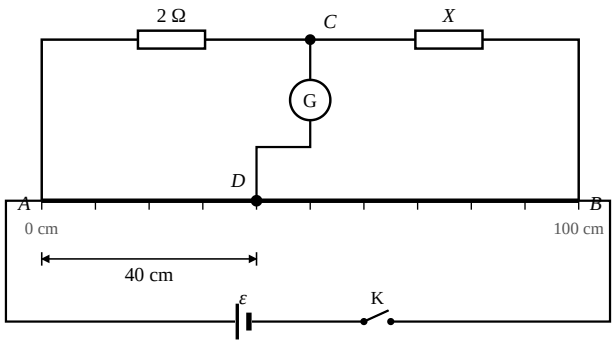


Figure 5

Question 11

State Ampère's circuital law. Using it, obtain an expression for the magnetic flux density B at a point X at a perpendicular distance r from a long straight conductor carrying a current I. *(Understand)* [3]

Question 12

(i) Two long straight parallel wires 10 cm apart carry currents of 5 A and 8 A in the same direction. Calculate the force per unit length between them and state whether it is attractive or repulsive. *(Apply)* [1]

(ii) A galvanometer of resistance 50 Ω gives full-scale deflection for a current of 2 mA. How can it be converted into a voltmeter of range 0–20 V? Find the resistance of the voltmeter. *(Apply)* [2]

Question 13

Using the arrangement of Young's double slit experiment shown in Figure 6, a teacher asked the following questions to a student. State whether the answer of the student is correct or incorrect. Correct the incorrect response with a brief explanation. *(Evaluate)*

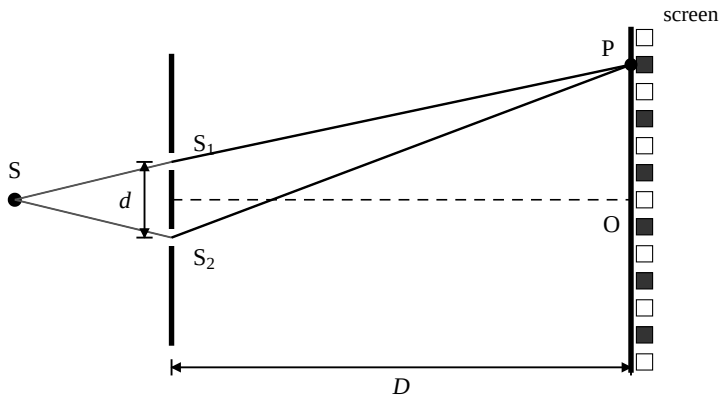


Figure 6

(i) **Teacher:** At the point P the path difference between the two interfering waves is $1{\cdot}25 \times 10^{-6}$ m and the wavelength of the light is 500 nm. Is P bright or dark? [1]
Student: Dark, because the path difference is an odd multiple of λ/2.

(ii) **Teacher:** The slits are 0·5 mm apart and the screen is 1·2 m away. What happens to the fringe width when the 600 nm light is replaced by 450 nm light? [1]
Student: It increases from 1·44 mm to 1·92 mm.

(iii) **Teacher:** The double slit is replaced by a single slit and the pattern is observed. How does the width of the central maximum compare with that of the other maxima? [1]
Student: The central maximum is narrower than the others.

Question 14

Using Huygens' wave theory and a labelled diagram, show that the angle of reflection is equal to the angle of incidence. *(Understand)* [3]

Question 15

For a prism of refracting angle A, show that the refractive index of its material is $n = \sin[(A + \delta_m)/2]/\sin(A/2)$, where δ_m is the angle of minimum deviation. *(Understand)* [3]

Question 16

(i) Draw a labelled ray diagram of an astronomical telescope when its final image lies at infinity. *(Understand)* [1]

(ii) An astronomical telescope has an objective of focal length 150 cm and an eyepiece of focal length 5 cm. Calculate (a) its magnifying power and the length of the tube in normal adjustment, and (b) the magnifying power when the final image is formed at the least distance of distinct vision (25 cm). *(Apply)* [2]

Question 17

(i) A monochromatic source emits light of wavelength 198 nm. Calculate the energy and the momentum of each photon. *(Apply)* [2]

(ii) Plot a labelled graph of the stopping potential V_s against the frequency ν of the incident radiation, and state how Planck's constant is obtained from it. *(Understand)* [1]

SECTION D – 15 MARKS

Question 18

(i)(a) A coil is connected to a 100 V d.c. supply and carries a current of 2 A. When it is connected to a 100 V, 50 Hz a.c. supply the current is 1 A. (1) Explain why the current is less with the a.c. supply. (2) Calculate the coefficient of self-inductance of the coil. *(Apply)* [3]

(b) A rod of length 0·4 m slides at a constant speed of 3 m s^{-1} over two parallel rails joined by a 5 Ω resistor, in a uniform magnetic field of 0·25 T perpendicular to the plane of the rails. Calculate (1) the induced emf and the current, and (2) the force needed to keep the rod moving at this speed. *(Apply)* [2]

OR

(ii)(a) A resistor *R*, an inductor *L* and a capacitor *C* are connected in series to an a.c. source. Using a phasor diagram, derive an expression for the impedance of the circuit. *(Understand)* [3]

(b) A series LCR circuit has R = 10 Ω, L = 5 mH and C = 20 μF and is connected to a 220 V a.c. source of variable frequency. Find the resonant frequency and the current at resonance. *(Apply)* [2]

Question 19

(i)(a) Using Bohr's postulates, derive an expression for the orbital velocity of the electron in the *n*th orbit of a hydrogen-like atom of atomic number *Z*. *(Understand)* [3]

(b) Hence show that the speed of the electron in the first orbit of hydrogen is about 1/137 of the speed of light. *(Apply)* [1]

(c) Calculate the shortest wavelength of the Balmer series of hydrogen. *(Apply)* [1]

OR

(ii)(a) Sketch a graph showing the variation of the binding energy per nucleon of a nucleus with its mass number. Mark on it the regions of nuclei that are (1) stable, (2) prone to fission, (3) prone to fusion. *(Understand)* [3]

(b) Calculate the energy released in the nuclear reaction $^{2}_{1}H + {}^{2}_{1}H \rightarrow {}^{4}_{2}He$ + energy. (Mass of $^{2}_{1}H$ = 2·014102 u, mass of $^{4}_{2}He$ = 4·002604 u, 1 u = 931·5 MeV/c^2) *(Apply)* [2]

Question 20

Ravi builds a small d.c. power supply for his school laboratory. He starts with a p–n junction diode and studies how the width of its depletion region changes with the bias. He then connects two such diodes and a centre-tapped transformer, in the 50 Hz mains circuit, to get a full-wave rectifier, and compares its output with that of a half-wave rectifier. Finally he uses a Zener diode to keep the output voltage constant.

(i) How does the thickness of the depletion region of a p–n junction change in forward bias and in reverse bias? *(Recall)* [1]

(ii) Draw a neat, labelled circuit diagram of the full-wave rectifier built by Ravi. *(Understand)* [1]

(iii) Show the input and the output waveforms for this rectifier. *(Understand)* [1]

(iv) What are the frequencies of the ripple in the output of the half-wave and of the full-wave rectifier for a 50 Hz input? *(Apply)* [1]

(v) A Zener diode of breakdown voltage 6·0 V is connected in the reverse direction across the output through a 300 Ω series resistor from a 12 V supply (no load connected). Find the current through the series resistor. *(Apply)* [1]

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